6.7 The Quaternion Group

WORKSHEET

The question

STEP III 1993, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Think of the pair (z_1, z_2) as standing for a "number" $z_1 + z_2\mathrm{j}$, where j is a new symbol with $\mathrm{j}^2 = -1$ and $\mathrm{j}w = w^*\mathrm{j}$ for complex w. Multiply $(z_1 + z_2\mathrm{j})(w_1 + w_2\mathrm{j})$ using these rules. Do you get the formula for $\circ$?

STAGE 1 Associativity via matrices

Covers: associativity

Ask yourself Checking $(a \circ b) \circ c = a \circ (b \circ c)$ directly is long. Is there a slicker way?

Pause and try Match (z_1, z_2) with the complex matrix $\begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix}$. What is the top row of the product of two such matrices? Matrix multiplication is associative.

STAGE 2 Not commutative

Covers: commutativity, and the four products

Ask yourself Try $(0,1) \circ (\mathrm{i},0)$ and $(\mathrm{i},0) \circ (0,1)$. Then put zeros into the definition to get the four special products.

STAGE 3 The group of eight

Covers: the set S_1

Ask yourself Use the four special products to show that S_1 is closed. What is the identity?

Pause and try Find each element's square. Which elements square to the identity, and which to $(-1, 0)$? Use that to find each order and inverse.

STAGE 4 A subgroup

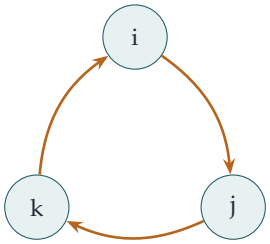
Covers: the subgroup and the matrix

Ask yourself Work out the powers of B. Then the powers of the given matrix. Match them up.

What's really going on

Beyond the complex numbers

The real numbers form a one-dimensional number system in which you can add, multiply and divide; the complex numbers form a two-dimensional one. Hamilton's quaternions are four-dimensional, and you can still divide, but multiplication is no longer commutative (Figure 6.7.1). In 1878 Ferdinand Frobenius proved that these three, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$ (for Hamilton), are the only finite-dimensional "division algebras" over the reals with associative multiplication. If you give up associativity but keep a size that multiplies, $|xy| = |x|\,|y|$, as it does for real and complex numbers and quaternions, there is exactly one more: the eight-dimensional *octonions*, discovered by John Graves and Arthur Cayley in the 1840s. Adolf Hurwitz proved in 1898 that these four are the only such systems. (Without the condition on size there are other, stranger non-associative division algebras, but a deep theorem of 1958, due to Raoul Bott, John Milnor and Michel Kervaire, shows that every finite-dimensional real division algebra has dimension 1, 2, 4 or 8.)



With the arrows: $\mathrm{ij} = \mathrm{k}$, $\mathrm{jk} = \mathrm{i}$, $\mathrm{ki} = \mathrm{j}$

Against them: $\mathrm{ji} = -\mathrm{k}$, $\mathrm{kj} = -\mathrm{i}$, $\mathrm{ik} = -\mathrm{j}$

and $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = -1$

Figure 6.7.1 Hamilton's multiplication rules. Multiplying two of i, j, k in the direction of the arrows gives the third; multiplying against the arrows gives minus the third. The eight elements $\pm 1, \pm \mathrm{i}, \pm \mathrm{j}, \pm \mathrm{k}$ form the quaternion group Q_8 of this question.

Rotations in three dimensions

The quaternions of modulus 1 describe rotations of three-dimensional space: the rotation through angle θ about a unit axis $\mathbf{u}$ corresponds to $q = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}(u_1\mathrm{i} + u_2\mathrm{j} + u_3\mathrm{k})$, acting on a vector $\mathbf{v}$ (written as $v_1\mathrm{i} + v_2\mathrm{j} + v_3\mathrm{k}$) by $\mathbf{v} \mapsto q\mathbf{v}q^{-1}$. Composing rotations is multiplying quaternions. This is more efficient and numerically stable than using matrices or angles, and it avoids "gimbal lock", a loss of one degree of freedom that the Apollo astronauts had to work around with their gyroscope-based navigation. The half-angle $\frac{\theta}{2}$ is also a clue to something deep: rotating by 360° gives $q = -1$, not 1, a fact that physicists see in the behaviour of electrons (spin).

Four squares

The modulus of a quaternion is multiplicative, $|pq| = |p||q|$, just like that of a complex number. Writing it out gives Euler's *four-square identity*: a product of two sums of four squares is itself a sum of four squares. That is the key step in Lagrange's theorem that every positive integer is a sum of four squares, which we prove in Section 7.8, the four-dimensional cousin of the two-squares result in Section 1.6.

Try this next

1 **Hamilton's rules.** Using the pairs, check that $C \circ B = D$ and $B \circ C = H$, that is, $\mathrm{ij} = \mathrm{k}$ and $\mathrm{ji} = -\mathrm{k}$.

2 **Only one element of order 2.** Show that $E = -1$ is the only element of order 2 in Q_8. Deduce that Q_8 is not isomorphic to the group of symmetries of a square (which has five elements of order 2).

3 **Four squares.** Show that $\det M(z_1, z_2) = |z_1|^2 + |z_2|^2$. Using $\det(M_1M_2) = \det M_1 \det M_2$, deduce that a product of two sums of four squares is a sum of four squares.

4 Subgroups. Find all the subgroups of Q_8. Show that every subgroup contains E unless it is $\{A\}$, and that Q_8 is the union of its three cyclic subgroups of order 4 (compare Section 6.6).

Further reading

- **John Conway & Derek Smith, *On Quaternions and Octonions*** (A. K. Peters). The geometry and arithmetic of quaternions and octonions. Advanced but fascinating.
- **John Baez, "The octonions"**, *Bulletin of the American Mathematical Society* 39 (2002); also freely available online. A wonderful survey; the introduction is readable now.
- **Andrew Hanson, *Visualizing Quaternions*** (Morgan Kaufmann). Quaternions for rotations, with applications to graphics.

The full guided solution is in Section 6.7:
https://beyondstep.keithmcnulty.org/ch06/quaternions.html